

Mohith Munagala

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PROFESSIONAL SUMMARY

Cloud, DevOps, and AI Infrastructure Engineer with 4+ years of experience designing, automating, and operating enterprise-grade cloud and on-prem platforms across AWS and Azure. Strong expertise in Docker and Kubernetes (EKS, AKS, and on-prem clusters) for deploying scalable, GPU-enabled AI and microservices workloads. Hands-on experience implementing GitOps-driven CI/CD pipelines using Argo CD, Azure DevOps, and Jenkins to deliver secure, automated, and version-controlled deployments. Proven background in building Large Language Model (LLM) platforms using Retrieval-Augmented Generation (RAG), integrating PostgreSQL and Neo4j Knowledge Graphs for enterprise AI chatbot and export control systems. Skilled in Infrastructure as Code (Terraform, Helm, CloudFormation), cloud security (IAM, KMS, Azure Key Vault), and observability using Prometheus, Grafana, and Application Insights. Recognized for optimizing GPU resource utilization, improving deployment efficiency, and delivering production-ready AI infrastructure aligned with enterprise compliance and governance standards.

EDUCATION

Eastern Illinois University – Charleston, Illinois

Aug 2023 – May 2025

MS in Computer Science

Relevant Coursework: Cloud Computing, DevOps, Infrastructure as Code (IaC), CI/CD, Kubernetes, Scripting, Cloud Architecture, Security, AI Models and Automation Tools.

TECHNICAL SKILLS

Cloud Platforms: AWS (EC2, S3, RDS, Lambda, VPC, IAM, KMS, CloudWatch, Route 53, API Gateway, SQS, EKS, DynamoDB, AWS Glue, Redshift), Microsoft Azure (App Services, AKS, Key Vault, Azure AD, ARM Templates, Service Principal)

AI & GPU Platforms: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG / Light RAG), Knowledge Graphs, Ollama, GPU-Accelerated Workloads, NVIDIA H200 GPU Infrastructure, AI Model Inference Deployment

Operating System: Amazon Linux/Unix, Ubuntu, Debian, Windows server

Infrastructure as Code: Terraform, AWS CloudFormation, Helm, AWS CDK, Azure Bicep, AWS CLI, Ansible, Puppet, Chef

CI/CD & GitOps: Argo CD, GitHub Actions, GitLab CI/CD, Jenkins, Azure DevOps, AWS Code Deploy, Maven, Nexus

Containerization & Orchestration: Docker, Kubernetes (EKS, AKS, On-Prem), OpenShift, Helm, Traefik Ingress Controller, Longhorn Distributed Storage

Monitoring & Logging: Prometheus, Grafana, Azure Application Insights, Amazon CloudWatch, Azure Monitor, Splunk, Datadog, JMeter, Locust

Security & Compliance: AWS IAM, Azure RBAC, KMS, Azure Key Vault, VPC Security Groups, S3 Bucket Policies, OAuth 2.0, Veracode, SonarQube, HIPAA, GDPR

Scripting & Automation: Python, Bash, PowerShell, YAML, JSON, Spring Boot, Java, JavaScript, Node.JS

Databases & Storage: PostgreSQL, MySQL, DynamoDB, Elasticsearch, Azure SQL, Neo4j (Graph Database)

Networking: VPC Architecture, NAT Gateway, Route Tables, DNS (Route 53), Load Balancers (ALB/NLB), Security Groups, SSL/TLS, Ingress Routing

APIs & Microservices: REST APIs, SOAP APIs, OAuth 2.0, AWS API Gateway, Lambda-based Microservices

DevOps Tools: GitOps, Blue/Green & Canary Deployments, Agile/Scrum, Cucumber, Runbook Automation, Infrastructure Automation

EXPERIENCE



Caterpillar
Cloud Engineer

Peoria, IL
Sep 2025 – Feb 2026

- Led infrastructure design and deployment of REDI 3.0, an enterprise AI chatbot platform built on GPU-enabled on-prem Kubernetes clusters powered by NVIDIA H200 GPUs to support large-scale LLM inference workloads.
- Architected and implemented Retrieval-Augmented Generation (Light RAG) pipeline integrating LLMs with

PostgreSQL and Neo4j Knowledge Graph to deliver context-aware, compliance-controlled AI responses.

- Deployed and managed LLM models using Ollama within GPU-backed environments to streamline model orchestration, inference serving, and version control.
- Designed and optimized GPU scheduling and resource allocation within Kubernetes, significantly improving model inference performance and compute efficiency.
- Engineered end-to-end AI platform using Dockerized microservices deployed via Kubernetes with autoscaling, self-healing, and high-availability configurations.
- Implemented GitOps-driven deployment strategy using Argo CD, enabling controlled, versioned, and automated rollouts across dev, QA, and production clusters.
- Built and configured GitHub self-hosted runners within on-prem infrastructure to support secure CI/CD pipelines for container builds and Kubernetes deployments.
- Built enterprise-grade Knowledge Graph in Neo4j to enforce export control policies and enhance semantic search capabilities within chatbot workflows.
- Designed secure ingress architecture using Traefik with SSL termination and traffic routing to expose AI services securely within enterprise network.
- Implemented distributed storage architecture using Longhorn to ensure persistent, highly available storage for AI workloads across cluster nodes.
- Established full-stack observability using Prometheus, Grafana, and Azure Application Insights to monitor GPU utilization, latency, API performance, and system reliability.
- Integrated Azure Service Principals, RBAC, and Azure Key Vault to enforce secure authentication, authorization, and secret management across AI services.
- Conducted performance benchmarking, capacity planning, and infrastructure optimization for high-throughput AI inference environments.
- Owned production stability, performing root cause analysis across Kubernetes networking, GPU drivers, container runtime, and database layers.



British Airways
Senior Cloud Engineer

New York, USA
June 2024 – Sep 2025

- Designed and developed the NDC platform using AWS EKS, Auto Scaling, and ALB/NLB to serve high-traffic flight bookings.
- Architected and provisioned infrastructure using Terraform, Helm, and AWS CloudFormation, enabling seamless deployments for Kubernetes workloads.
- Built secure VPC architectures with IAM, KMS, route tables, NAT gateways, and subnets to meet airline security and compliance standards.
- Developed and optimized CI/CD pipelines using GitHub Actions and Jenkins to automate the build, scan, and deployment of over 40 microservices.
- Designed and implemented RESTful APIs for airline offer and order services using AWS API Gateway and integrated them with AWS Lambda for serverless compute.
- Developed and deployed secure SOAP API endpoints for IATA schema validations, interfacing with external airline systems, which improved data accuracy and compliance with industry standards.
- Developed and maintained Python scripts using Boto3 to automate EKS node group scaling, perform S3 access audits, and streamline log shipping to CloudWatch and S3 buckets for the NDC microservices platform, enhancing operational efficiency and reducing manual intervention.
- Implemented blue/green and canary deployments using Kubernetes and Helm charts to support zero-downtime releases, which increased deployment reliability and minimized service disruptions.
- Configured observability solutions using Prometheus, Grafana, CloudWatch, and Splunk, ensuring high availability and proactive monitoring, which improved system uptime and incident response times.
- Implemented and monitored containerization using Docker and Kubernetes for scalable deployments in the NDC platform, improving reliability and reducing operational overhead.
- Conducted performance testing using JMeter and Locust, tuning autoscaling configurations for dynamic workloads, which optimized resource utilization and improved application performance under varying loads.
- Led disaster recovery and high-availability planning across Multi-AZ with Route 53 and CloudFormation-based failover plans, ensuring minimal downtime and enhanced system reliability.
- Worked with QA teams to validate API contract testing using Postman, SoapUI, and custom Python scripts,

leading to more accurate testing outcomes and reduced defects.

- Participated in Agile ceremonies, sprint planning, and release coordination with NDC product owners and QA leads, facilitating smoother project execution and improved team collaboration.
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TCS
Cloud Engineer

Hyderabad, India
June 2021 – July 2023

- Provisioned and maintained secure and scalable AWS infrastructure using EC2, VPC, S3, RDS, Lambda, and IAM to support large-scale financial applications
- Automated infrastructure deployments using Terraform and AWS CloudFormation across development, UAT, and production environments.
- Developed and improved CI/CD pipelines using GitLab CI and AWS Code Deploy to automate build, test, and release cycles for containerized services, increasing deployment efficiency.
- Deployed microservices on Amazon EKS using Helm charts and managed Docker image versions via private container registries.
- Led Docker containerization and Kubernetes orchestration for financial services, streamlining deployments, enabling elasticity, and optimizing CI/CD workflows.
- Managed versioning, authentication, and schema validation for REST and SOAP APIs using Terraform and CloudFormation templates.
- Developed REST APIs using AWS Lambda and API Gateway to support real-time payment workflows and third-party integrations.
- Wrote Python and Bash scripts for EC2 instance automation, AMI backups, log cleanup, and CloudWatch alarm integration.
- Designed and implemented IAM policies, S3 bucket policies, and KMS encryption strategies to meet internal security and compliance requirements.
- Configured Prometheus and Grafana dashboards along with Amazon CloudWatch alerts for proactive infrastructure and application monitoring.
- Supported serverless workloads using AWS Lambda, SQS, and API Gateway to enable real-time processing for payment transaction systems.
- Conducted root cause analysis and incident response using AWS CloudTrail, VPC Flow Logs, and Splunk logs.
- Participated in daily Agile ceremonies, collaborated with cross-functional DevSecOps teams, and delivered sprint-based cloud deliverables.
- Maintained detailed runbooks, environment documentation, and automation repositories in Git for future reference and team onboarding.
- Developed and enforced cloud security controls and audit policies aligned with HIPAA, GDPR, and internal governance frameworks, collaborating with InfoSec and QA teams for periodic compliance validation.

CERTIFICATIONS

- AWS Certified Solution Architect
- Microsoft Azure Fundamentals
- AWS Certified Cloud Practitioner
- Certified Kubernetes
- Certified Terraform